WORKED SOLUTION

STEP II 1995, Question 9

Mechanics • Friction and Equilibrium

STEP II 1995, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

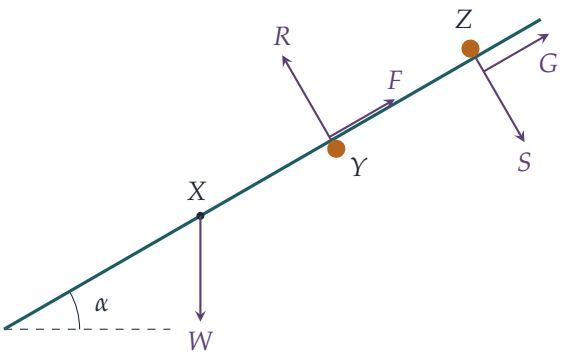
step.maths.org/questions/95-s2-q9

The key idea

The rod lies along a line of greatest slope, so it is natural to resolve along the rod and perpendicular to it. The bar the rod rests on pushes it up, away from the plane; the bar it passes under can only push it down. Taking moments about Y shows where the centre must be, and then the least μ comes from making both frictional forces act up the slope at their limiting values.

Solution

We model the rod as a uniform rigid rod of weight W and the bars as fixed thin bars, each giving a normal reaction perpendicular to the rod and a friction force along it. Since the rod is perpendicular to both horizontal bars, it lies along a line of greatest slope of their plane, and so it is inclined at α to the horizontal. Let R be the reaction at Y (pushing the rod away from the plane, since the rod lies on top of this bar) and S the reaction at Z (pushing the rod towards the plane, since the rod is underneath this bar), with $R, S \geq 0$. Let F and G be the friction forces at Y and Z, taken as positive up the slope.



Where the centre must be

Resolving perpendicular to the rod and taking moments about Y (distances measured along the rod):

$$R = S + W\cos\alpha,$$
$$S\,d = W\cos\alpha \cdot XY.$$

The moment of S about Y must be balanced by the moment of the weight component $W\cos\alpha$, which acts towards the plane just as S does. Two parallel forces in the same direction can only have cancelling moments about Y if they act on opposite sides of Y. Since Z is up the slope from Y, the centre X must be at or below Y, and then

$$XZ = XY + YZ, \qquad \text{so} \qquad l = XY + d \geq d.$$

(If X were between Y and Z, or beyond Z, the moment equation would force $S < 0$: the bar at Z would have to pull the rod, which it cannot do, and the rod would rotate off it.) Equality $l = d$ is the borderline case in which X is at Y and $S = 0$.

Ask yourself The paper says $l \leq d$. Why do we get the opposite inequality?

The inequality in the paper is a misprint: the question itself goes on to take $l = 2d$, which is only possible if $l \geq d$. The argument above shows that $l \geq d$ is what equilibrium requires, and that is what we use.

The least value of μ

With $XY = l - d$ the two equations give

$$S = \frac{W\cos\alpha\,(l-d)}{d}, \qquad R = \frac{W\cos\alpha\,l}{d}, \qquad \text{so} \qquad R + S = \frac{W\cos\alpha\,(2l-d)}{d}.$$

Resolving along the rod, friction must hold up the component of the weight down the slope:

$$F + G = W\sin\alpha.$$

The friction laws require $|F| \leq \mu R$ and $|G| \leq \mu S$, so $F + G \leq \mu(R+S)$. Hence equilibrium needs

$$W\sin\alpha \leq \mu\,\frac{W\cos\alpha\,(2l-d)}{d}, \qquad \text{that is,} \qquad \mu \geq \frac{d\tan\alpha}{2l-d}.$$

Conversely, if μ equals this value, then $F = \mu R$ and $G = \mu S$ (both up the slope, both limiting) satisfy every equation, so the rod can be in equilibrium. The least possible value is therefore

$$\mu_{\min} = \frac{d\tan\alpha}{2l-d}.$$

When $l = 2d$ this becomes $\mu_{\min} = \dfrac{d\tan\alpha}{3d} = \frac{1}{3}\tan\alpha$, so $\mu \geq \frac{1}{3}\tan\alpha$, as required.

Checks

- When $l = d$ the centre is over the lower bar, $S = 0$, and we get $\mu \geq \tan\alpha$: the familiar condition for a single particle resting on a rough slope.
- As $l \to \infty$ the reactions grow without limit, so very little friction is needed, and indeed $\mu_{\min} \to 0$.
- Numerically, with $\alpha = 0.7$ and $l = 2d$, the formula gives $0.28076\ldots$, which is $\frac{1}{3}\tan 0.7$. A computer also checked the force and moment equations about random points, and that equilibrium holds just above $\mu_{\min}$ and fails just below it.

Common slips

- Taking both reactions in the same direction. The bar the rod lies under can only push it towards the plane.
- Assuming friction is limiting at only one bar. To make μ as small as possible, both bars must provide their full friction, up the slope.

Beyond STEP

This is a small problem in statics with inequality constraints: the contact forces must satisfy linear equations together with the friction inequalities, and the least μ is the point at which the feasible set just becomes non-empty. The same structure, linear equations with linear inequalities, underlies linear programming and the analysis of frictional contact in engineering. For more on rigid-body statics and friction, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

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